

A counterexample to the convex-block difference-basis problem

arXiv:1711.01340, final Problem 7

June 14, 2026

Source question

The source paper asks the following final problem.

Let X be a Banach space with a convex block homogeneous basis (e_i) . Define

$$d_1 = e_1, \quad d_{i+1} = e_{i+1} - e_i \quad (i \geq 1).$$

Is X , endowed with the basis (d_i) , submultiplicative?

Problem 5. Find a Banach space X with an unconditional basis $(x_i)_i$ that is not equivalent to the unit vector basis of c_0 so that there exist a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to the unitization of X .

Remark 8.9. The unitization of James space $\mathbb{R}e_\omega \oplus J$ may be viewed as the space of all scalar sequences of bounded quadratic variation and similarly for every Banach space X with a subsymmetric sequence the space $\mathbb{R}e_\omega \oplus J(X)$ is the space of all scalar sequences of bounded X -variation. One may consider for any such X and any linearly ordered set I the space $V_X(I)$ of all functions $f : I \rightarrow \mathbb{R}$ of bounded X -variation. The norm on such a space is submultiplicative. The spaces $V_X[0, 1]$ were introduced in [AMP]. By very carefully combining the method of the present paper with the method from [MPZ] it is conceivable that one may obtain for every countable well ordered set I a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to $V_X(I)$.

Problem 6. Let X be a Banach space with a subsymmetric basis. For what ordered sets I does there exist a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to $V_X(I)$?

Remark 8.10. In [AMS] the notion of a convex block homogeneous sequence is introduced, i.e., a sequence that is equivalent to its convex block sequences. Known examples of such sequences are constructed using a space X with a subsymmetric basis $(x_i)_i$. The bases $(e_i)_i$ of $J(X)$ and $(v_i)_i$ of $\mathcal{J}_*(X)$ are both convex block homogenous. In either case, the difference basis is submultiplicative.

Problem 7. Let X be a Banach space with a convex block homogeneous basis $(e_i)_i$ and set $d_1 = e_1$ and for $i \in \mathbb{N}$ $d_{i+1} = e_{i+1} - e_i$. Is X endowed with $(d_i)_i$ submultiplicative?

Counterexample

Theorem. Problem 7 of arXiv:1711.01340 has a negative answer as printed. In ℓ_2 , the usual unit vector basis is convex block homogeneous, but the difference sequence

$$d_1 = e_1, \quad d_{i+1} = e_{i+1} - e_i$$

does not satisfy any uniform submultiplicative estimate for finite coordinate vectors.

Proof. The usual unit vector basis of ℓ_2 is convex block homogeneous: every normalized block sequence is orthonormal, hence isometric to the original basis. In particular, every normalized convex block sequence has the same property.

For a finitely supported scalar sequence $a = (a_i)$, with $a_i = 0$ after the support, one has

$$\left\| \sum_i a_i d_i \right\|_2^2 = \sum_i |a_i - a_{i+1}|^2.$$

Fix $N \geq 2$ and set

$$a_i = \begin{cases} i/\sqrt{N}, & 1 \leq i \leq N, \\ (2N - i)/\sqrt{N}, & N < i \leq 2N, \\ 0, & i > 2N. \end{cases}$$

Then

$$\left\| \sum_i a_i d_i \right\|_2^2 = \frac{2N - 1}{N} < 2.$$

On the other hand, applying the same coordinate vector in both factors would force control of

$$\left\| \sum_i a_i^2 d_i \right\|_2^2 = \sum_i |a_i^2 - a_{i+1}^2|^2.$$

Already the increasing side gives

$$\left\| \sum_i a_i^2 d_i \right\|_2^2 \geq \frac{1}{N^2} \sum_{i=1}^{N-1} ((i+1)^2 - i^2)^2 = \frac{1}{N^2} \sum_{i=1}^{N-1} (2i+1)^2,$$

which tends to infinity as $N \rightarrow \infty$.

Thus no constant C can satisfy

$$\left\| \sum_i a_i b_i d_i \right\|_2 \leq C \left\| \sum_i a_i d_i \right\|_2 \left\| \sum_i b_i d_i \right\|_2$$

for all finitely supported coefficient vectors a, b , since taking $b = a$ gives a bounded right-hand side and an unbounded left-hand side. $\square$

Remark. *The argument answers the literal printed formulation. If the intended question silently assumes that (d_i) is already a Schauder basis, or imposes another additional hypothesis excluding the classical Hilbert basis, this packet should be read as a wording counterexample rather than as a solution of that strengthened variant.*

Verification and provenance

The finite-dimensional calculation is checked independently in the bundled verification script. The packet README records the duplicate and bounded novelty checks performed on 2026-06-14. Local run indexes contained no prior packet for this exact convex-block difference-basis problem; web searches found only the source paper context, not a later answer.